Masonic Cancer Center
University of Minnesota

Alcohol and Cancer

Silvia Balbo

Professor
Division of Environmental Health Sciences
School of Public Health
Co-leader of the Carcinogenesis and Chemoprevention Program

NCI Comprehensive Cancer Center
A Cancer Center Designated by the National Cancer Institute

Alcohol is a carcinogen

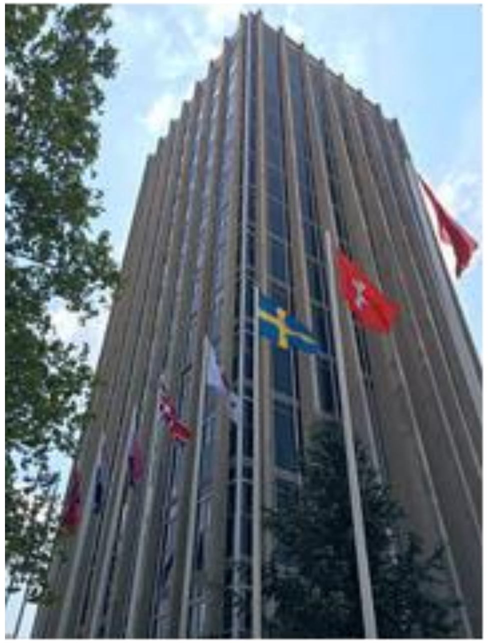
- It is classified as a Group 1 human carcinogen by IARC.
- Risk factor for cancers of the upper aerodigestive tract, liver, colon, and breast.
- Responsible for around 5-10% of all cancers in western countries.
- The risk does not seem to change depending on the beverage

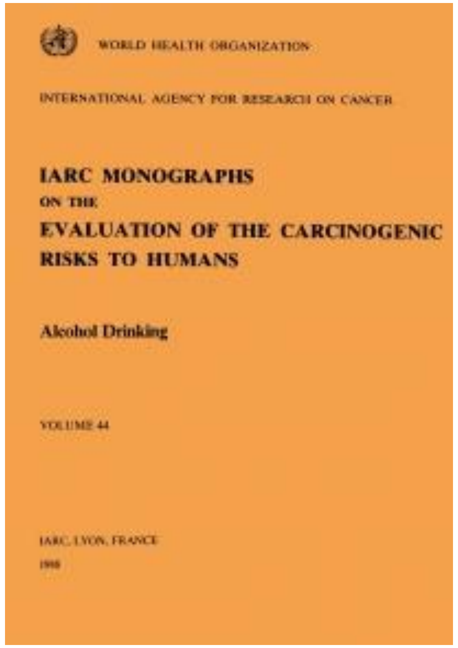
Over **20,000 people die** from **alcohol-related cancers** each year in the U.S.*

*During 2020-2021.

cdc.gov/alcohol

Alcohol is a carcinogen





WORLD HEALTH ORGANIZATION

INTERNATIONAL AGENCY FOR RESEARCH ON CANCER

IARC MONOGRAPHS
ON THE
EVALUATION OF THE CARCINOGENIC
RISKS TO HUMANS

Alcohol Drinking

VOLUME 44

IARC, LYON, FRANCE
1988

International Agency for Research on Cancer

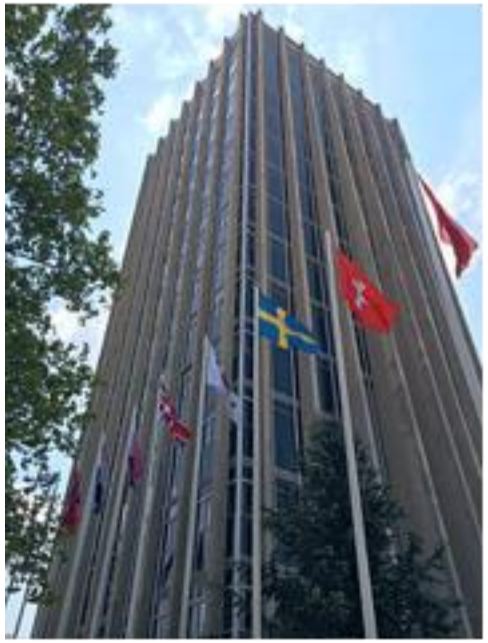
World Health Organization

IARC

Alcohol was classified as a Group 1 human carcinogen by IARC already in 1988.

The classification was based on evidence for an association with cancers of the oral cavity, larynx, pharynx, esophagus, and liver.

Alcohol is a carcinogen



International Agency for Research on Cancer

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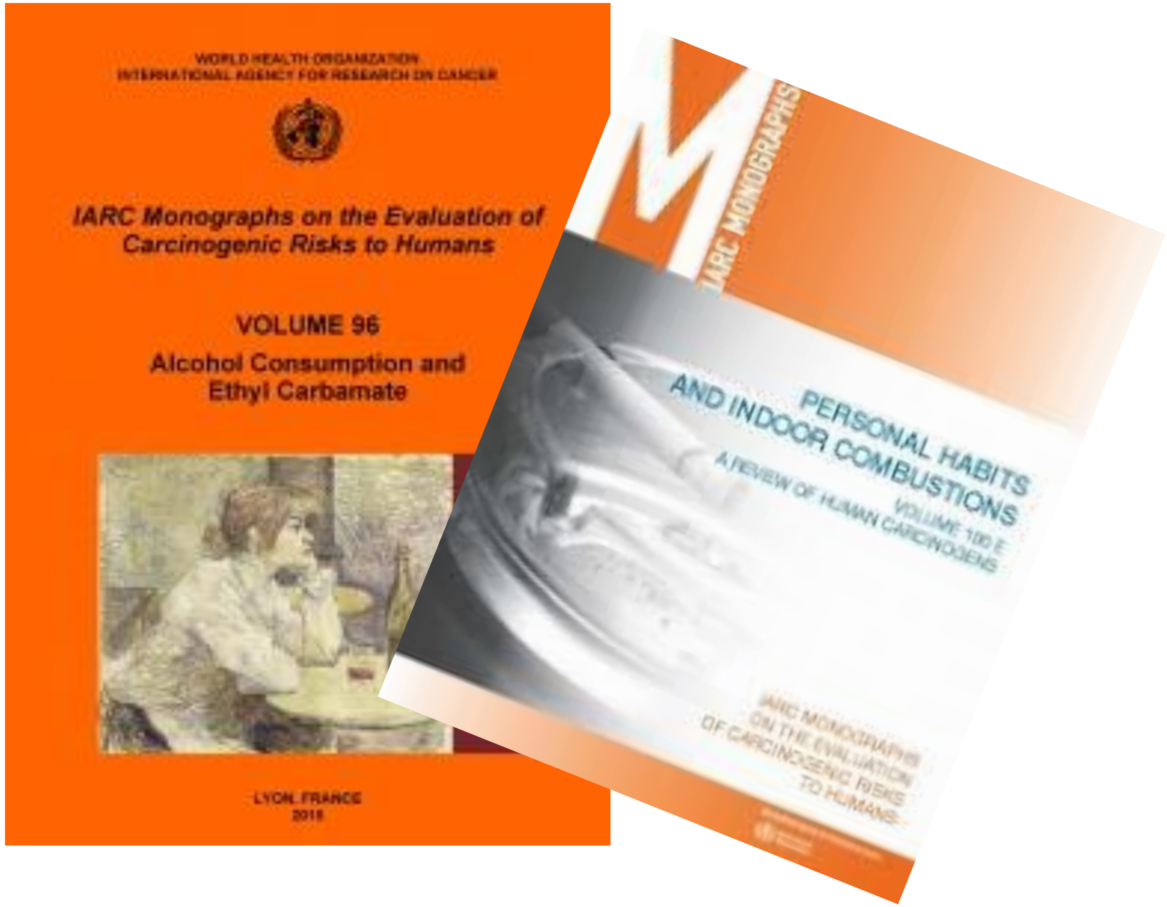
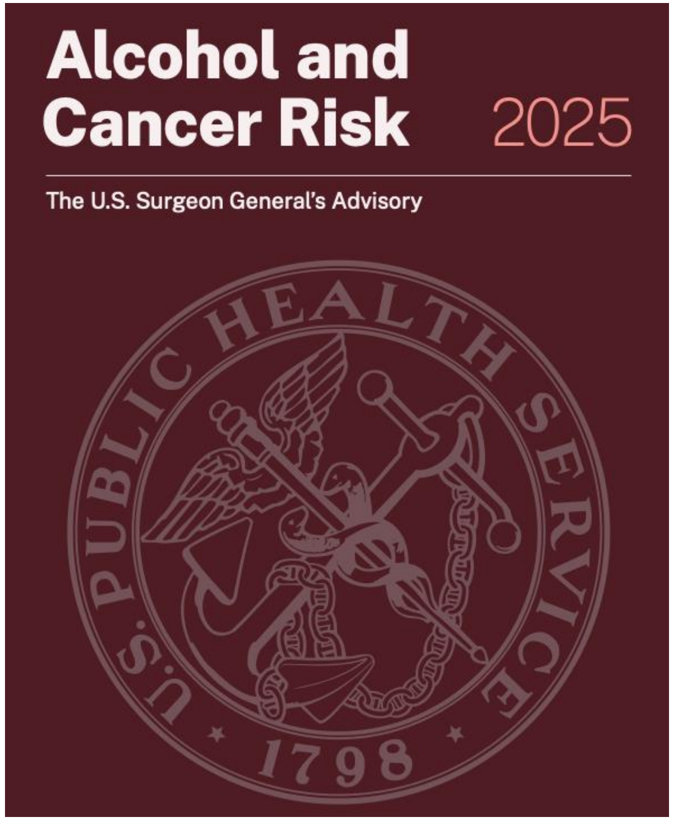
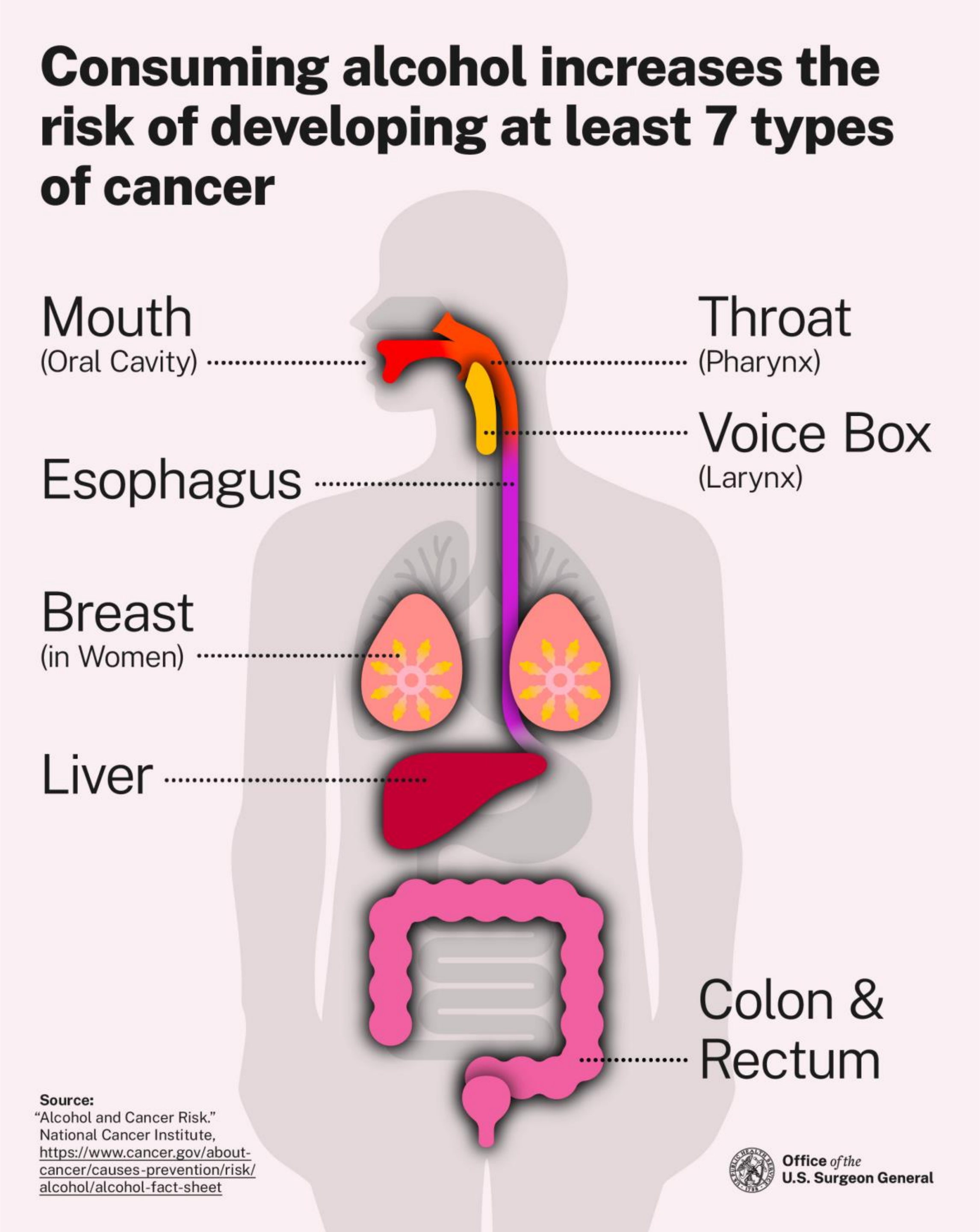
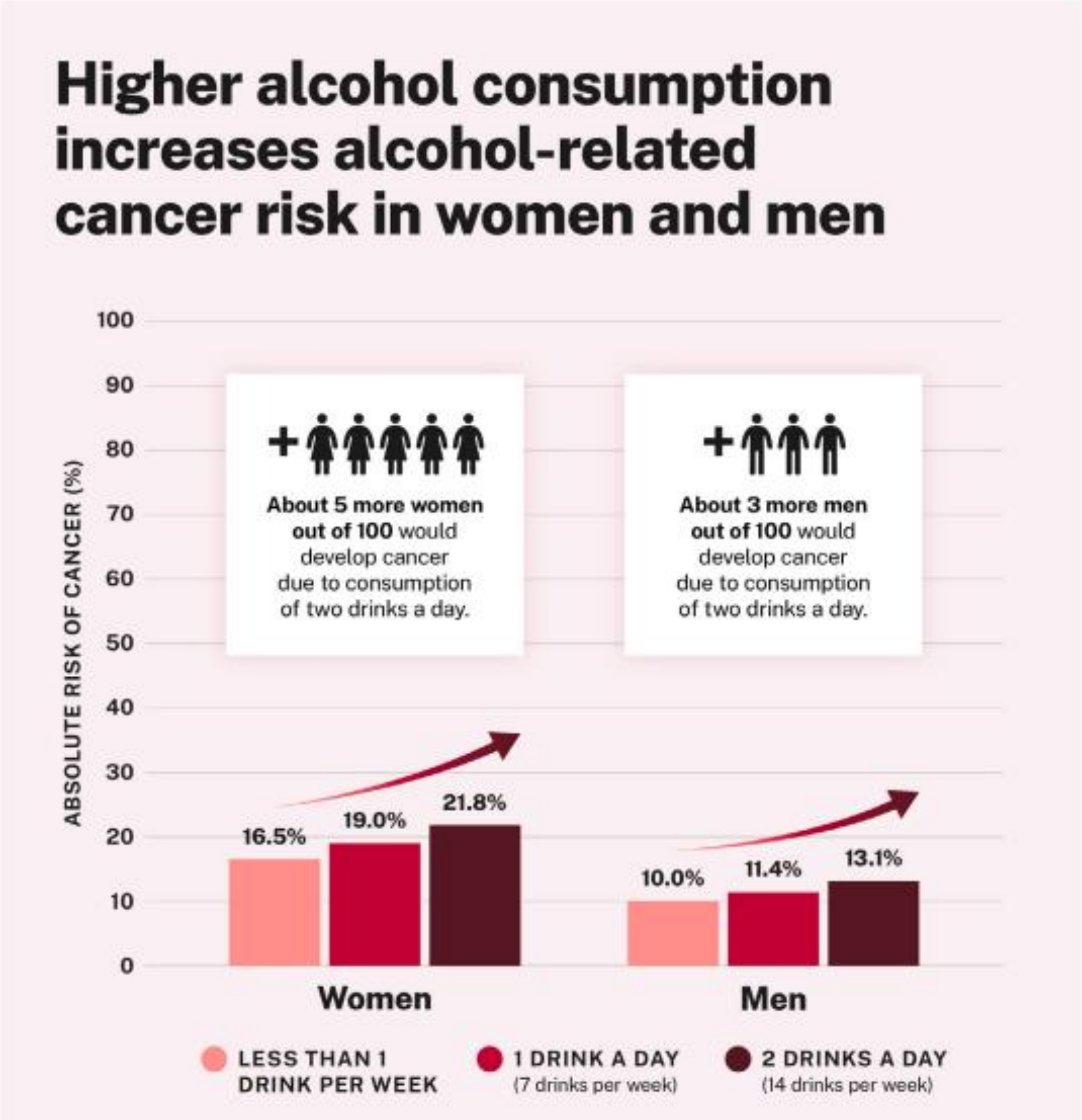


The classification was re-evaluated in 2012, and the associations with breast and colorectal cancer were added.

Alcohol and Cancer

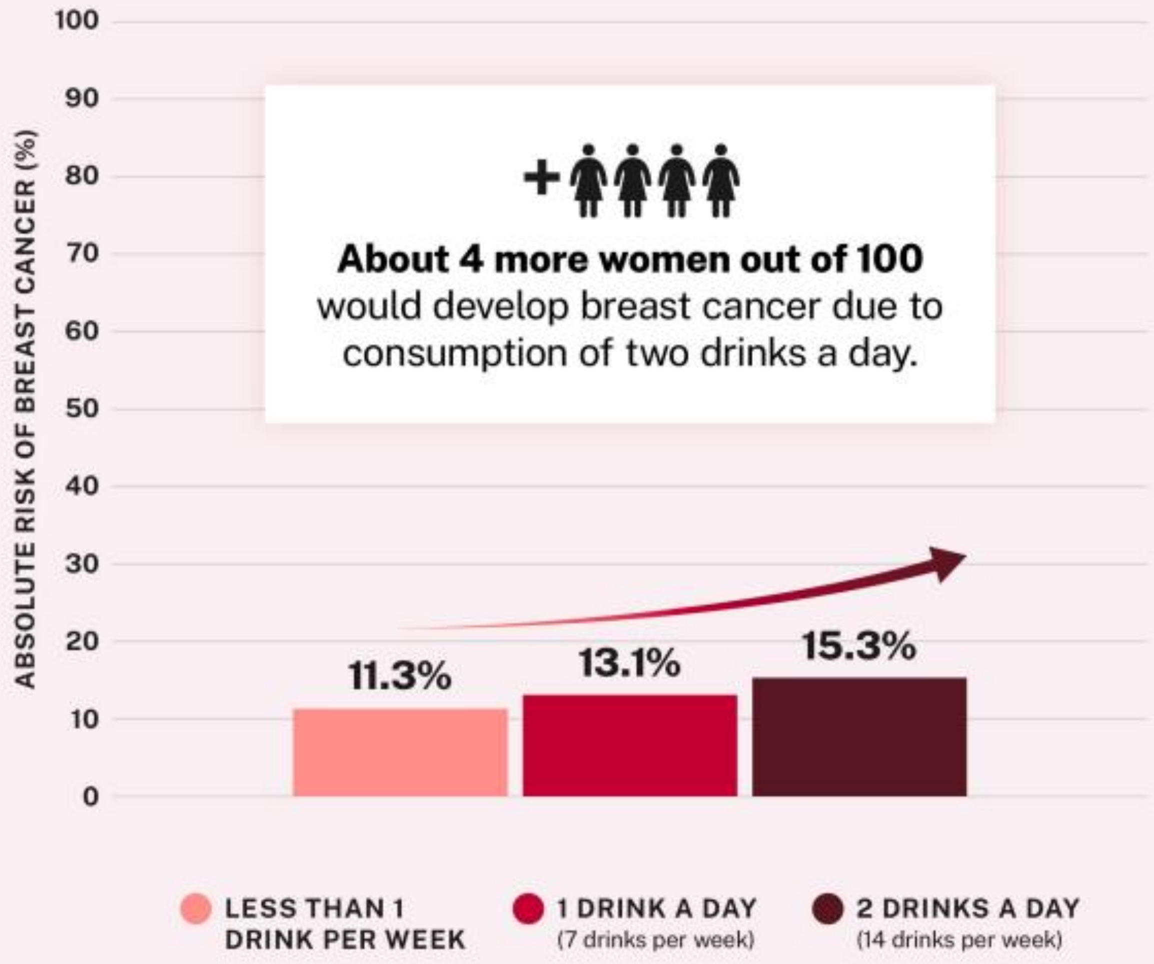






Higher alcohol consumption increases alcohol-related cancer risk in women and men
100
90
80
70
60
50
40
30
20
10
0
ABSOLUTE RISK OF CANCER (%)
About 5 more women out of 100 would develop cancer due to consumption of two drinks a day.
About 3 more men out of 100 would develop cancer due to consumption of two drinks a day.
16.5%
19.0%
21.8%
10.0%
11.4%
13.1%
Women
Men
LESS THAN 1 DRINK PER WEEK
1 DRINK A DAY (7 drinks per week)
2 DRINKS A DAY (14 drinks per week)

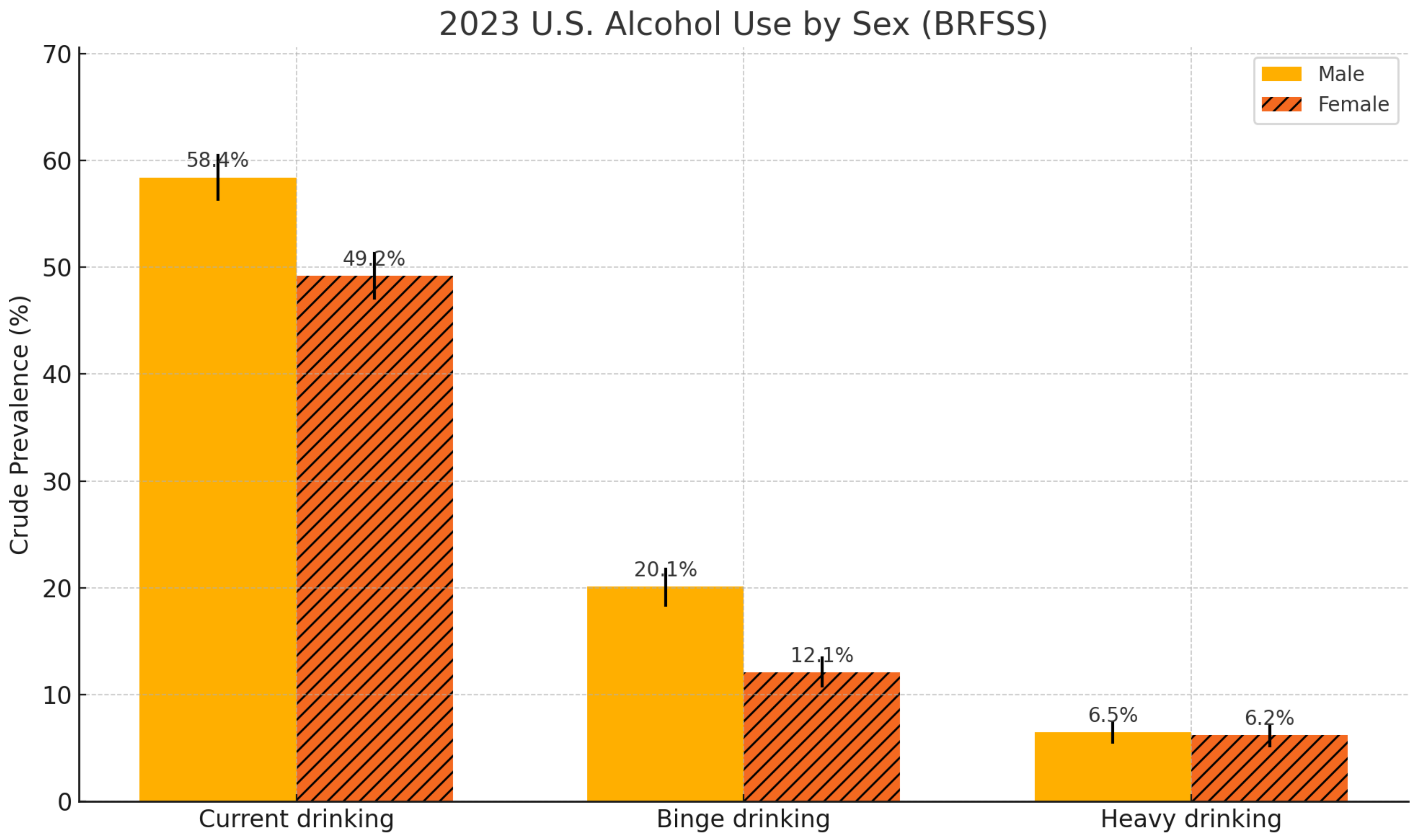
Higher alcohol consumption increases breast cancer risk in women



Alcohol-Attributable Deaths Due to Excessive Alcohol Use, Annual Average for United States, 2020-2021, All Ages

	Overall	Males	Females
Cancer			
Cancer, breast (females only)	2,594	x	2,594
Cancer, colorectal	2,832	2,453	379
Cancer, esophageal *	1,385	1,088	297
Cancer, laryngeal	687	621	66
Cancer, liver	5,109	4,683	426
Cancer, oral cavity and pharyngeal	2,916	2,494	422

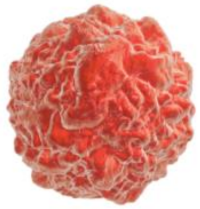
Behavioral Risk Factor Surveillance System



The mechanisms explaining Alcohol-related cancer risk remain unclear

Alcohol

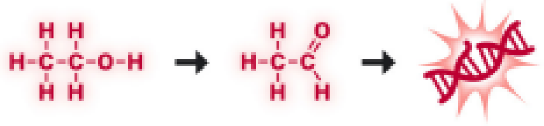
Cancer



Mechanisms of Cancer Formation

MECHANISM A

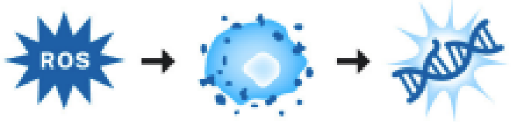
Alcohol breaks down into **acetaldehyde** which damages DNA in multiple ways, causing an increased risk of cancer.



Ethanol (Pure alcohol) → Acetaldehyde → DNA damage

MECHANISM B

Alcohol induces **oxidative stress**, increasing the risk of cancer by damaging DNA, proteins, and cells and increasing inflammation.



Reactive oxygen species → Oxydative stress → DNA, proteins, and lipids damage

MECHANISM C

Alcohol alters **levels of multiple hormones**, including estrogen, which can increase breast cancer risk.

Increases estrogen → Alters breast tissue → DNA damage

MECHANISM D

Alcohol leads to greater absorption of **carcinogens.**

Carcinogens dissolve in alcohol + Alcohol alters cells in mouth and throat → Carcinogens more easily absorbed

Mechanisms of Cancer Formation

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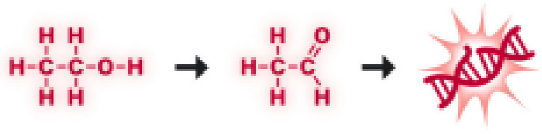

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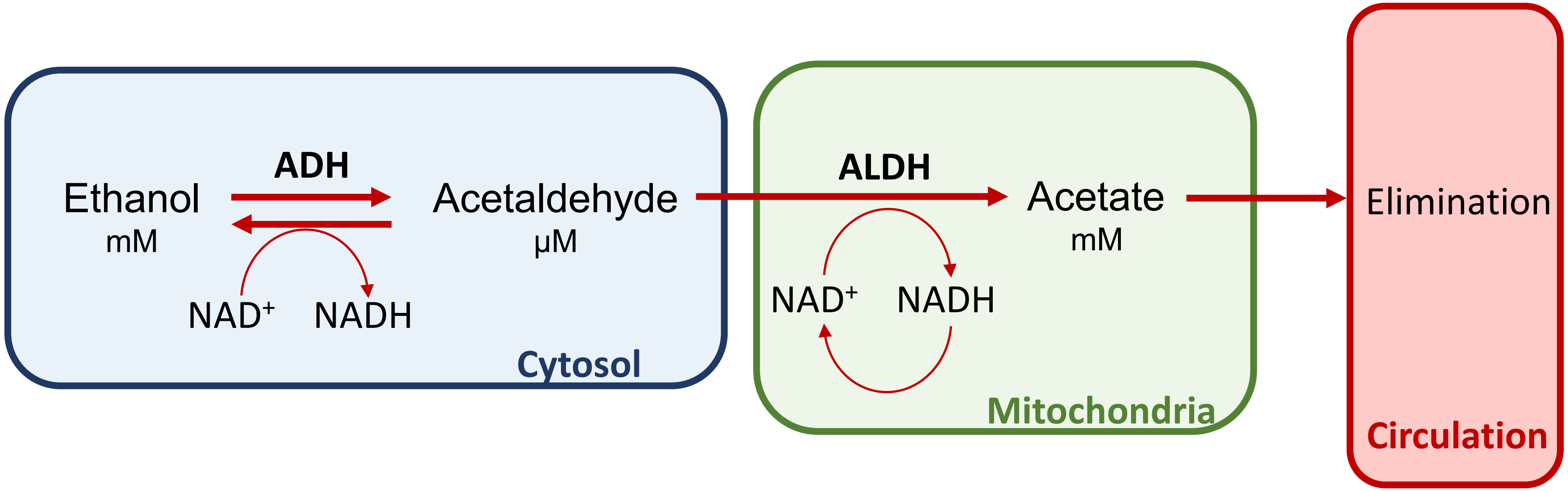
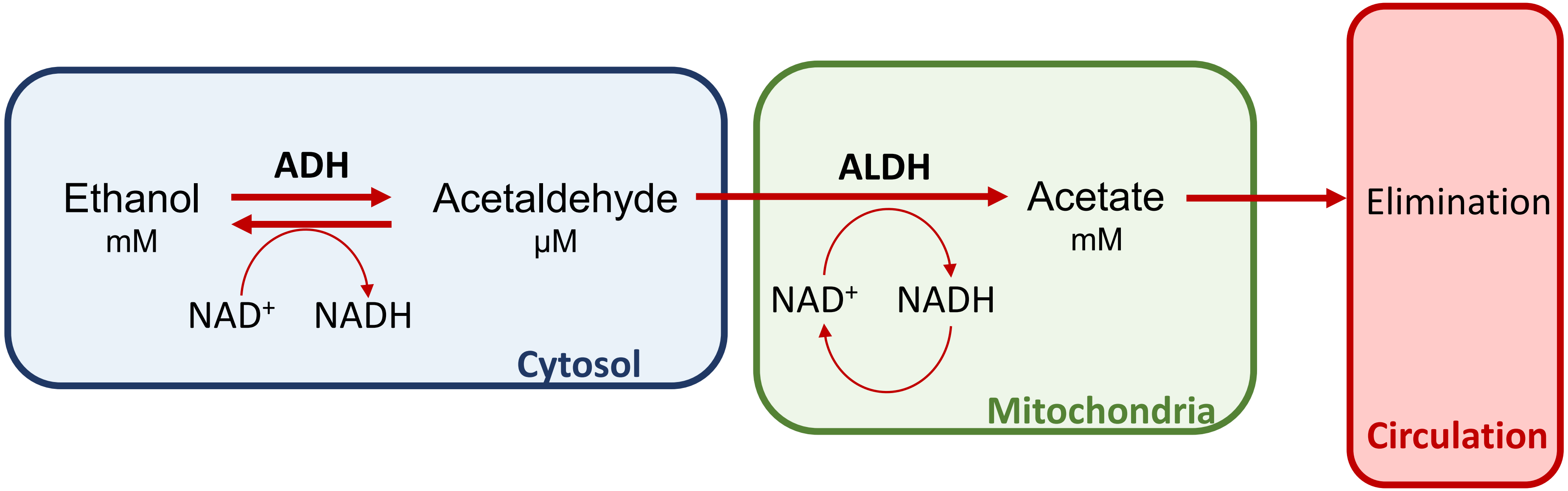
Acetaldehyde



Acetaldehyde



*ALDH2*1/2** heterozygotes:

- ~10% residual activity for this enzyme
- increased risk for head and neck cancer when drinking alcohol

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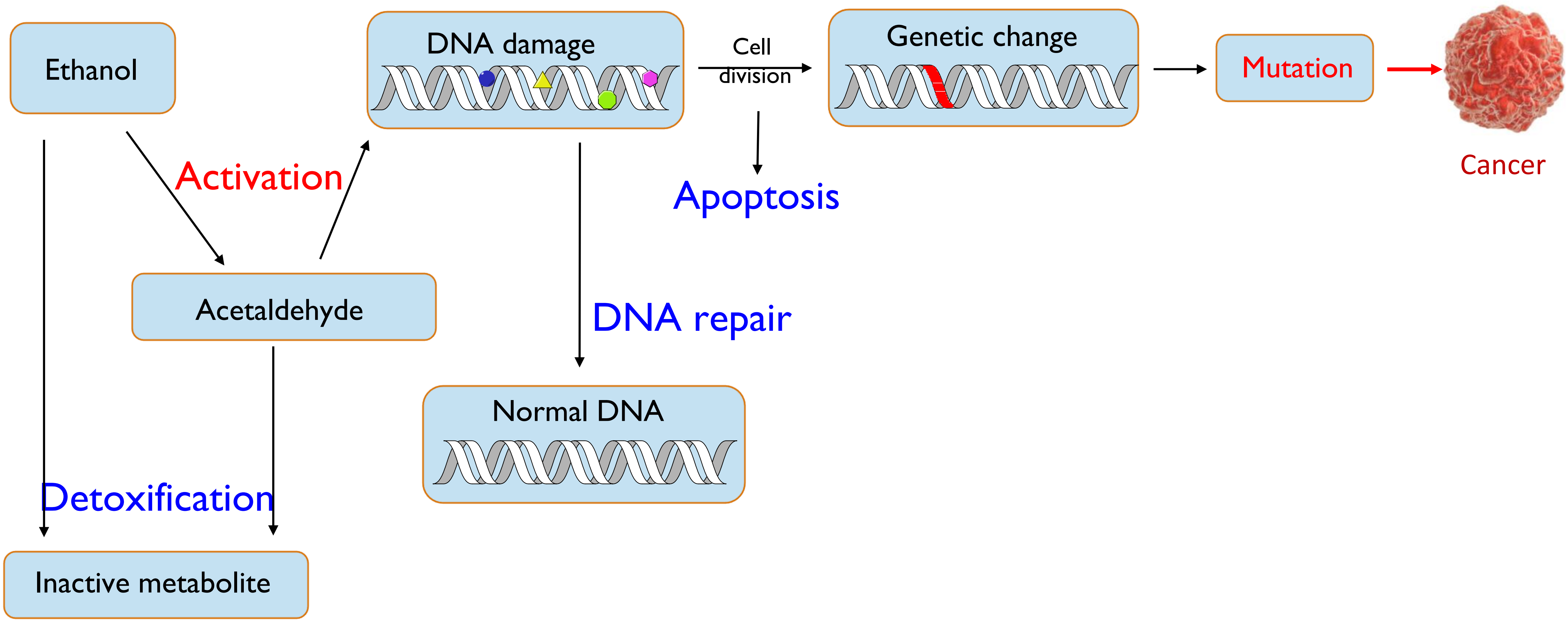
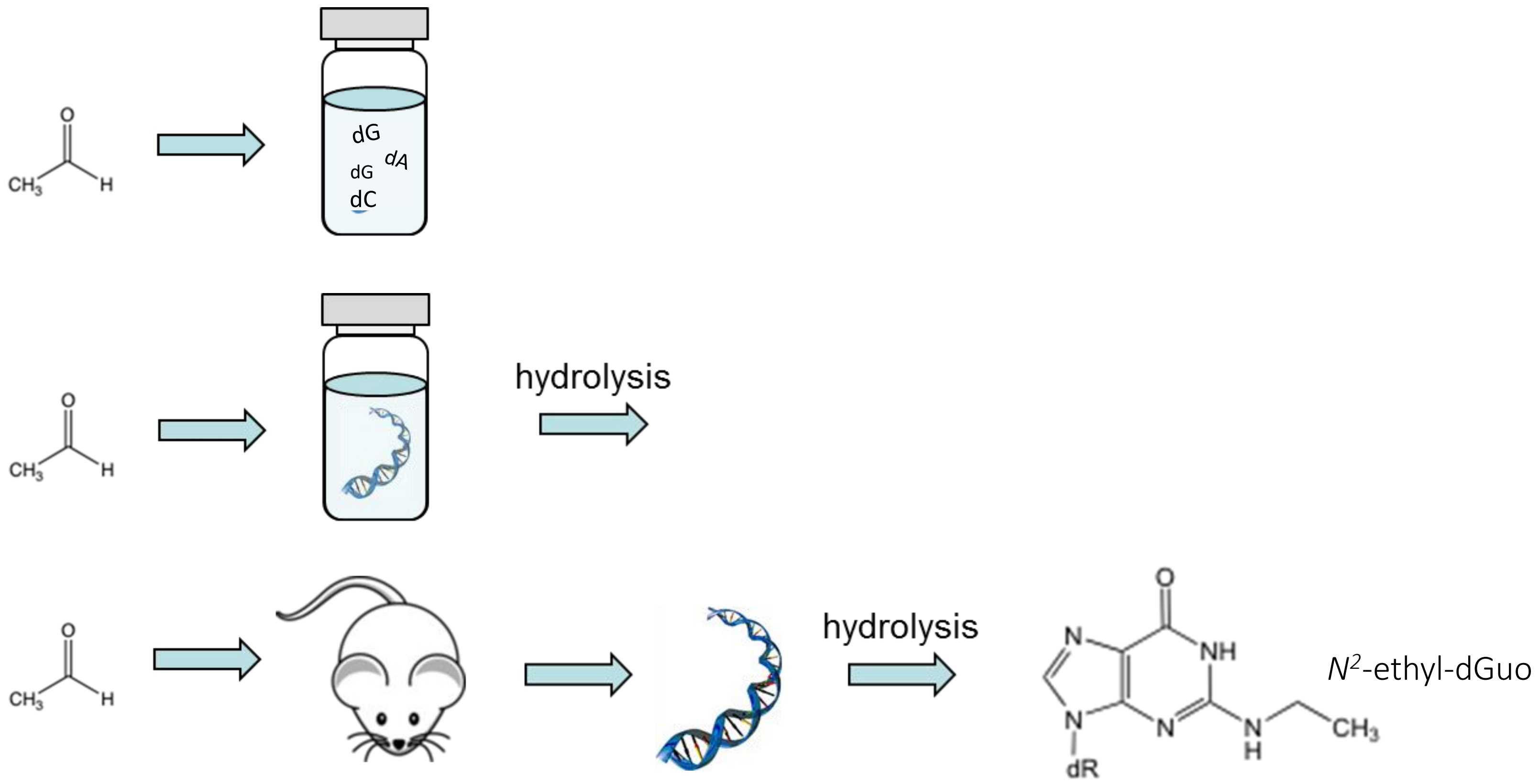
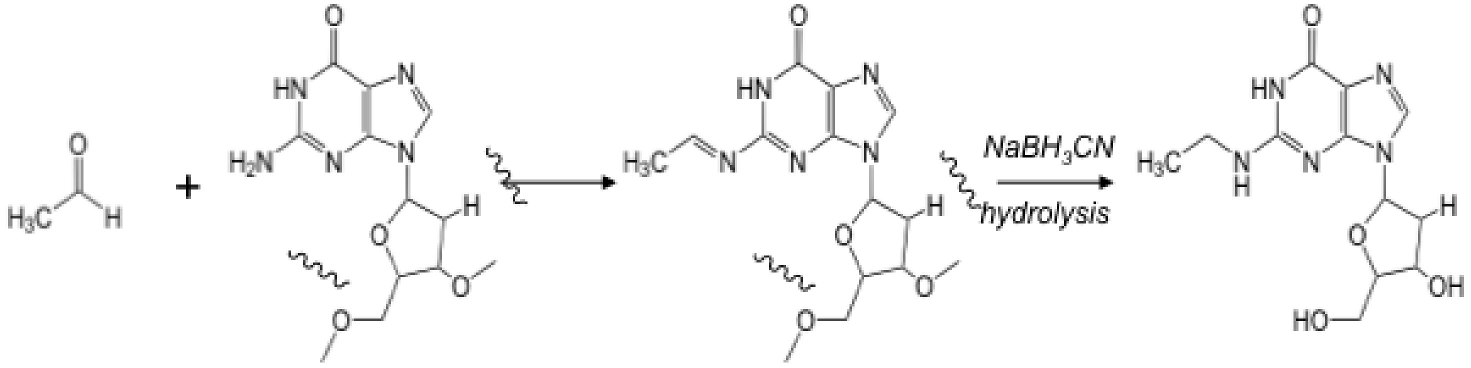
Acetaldehyde-derived DNA damage



Acetaldehyde-derived DNA Adducts

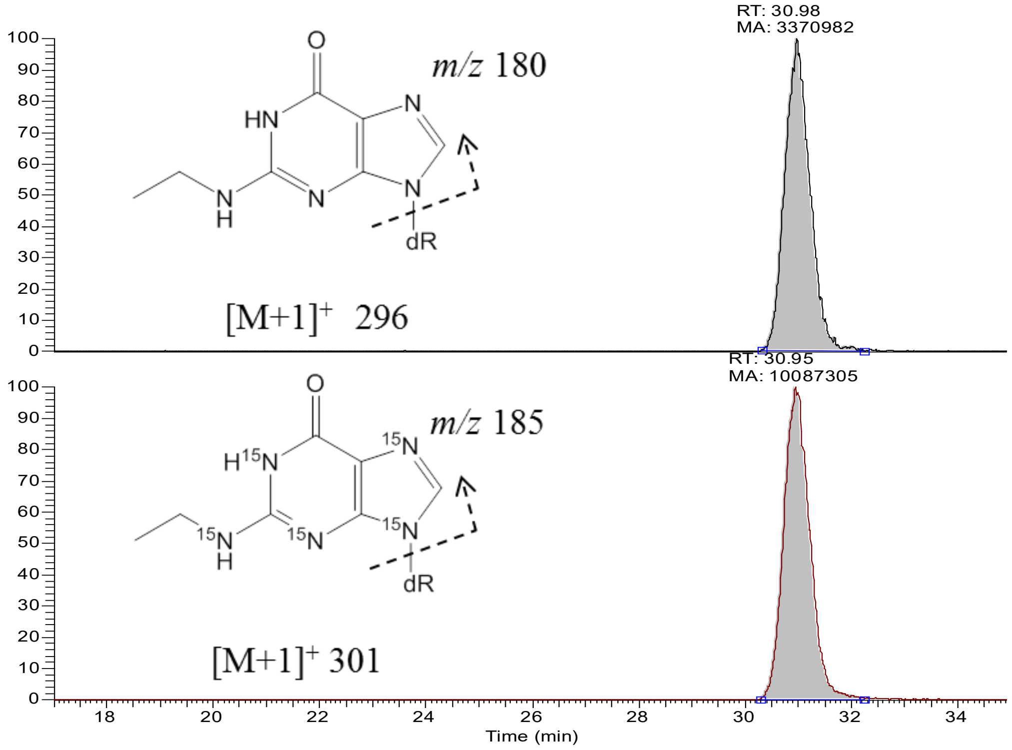


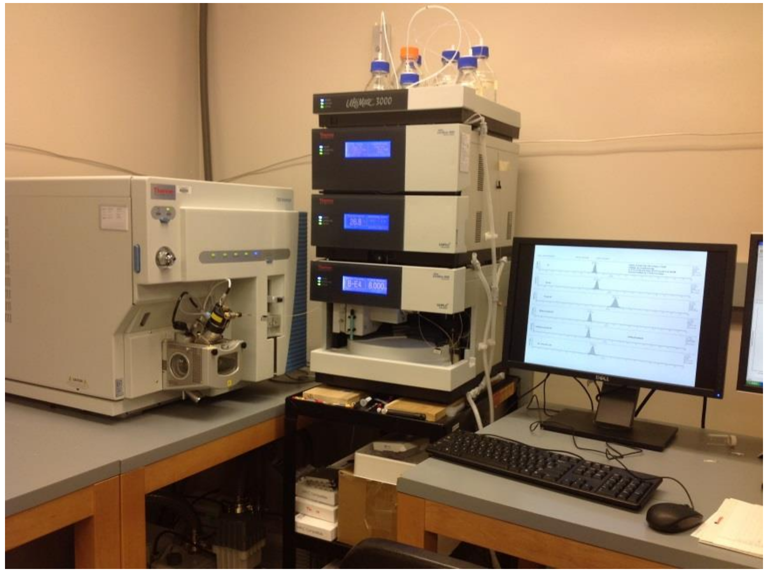


deoxyguanosine

N^2-ethylidene-dG

N^2-ethyl-dG

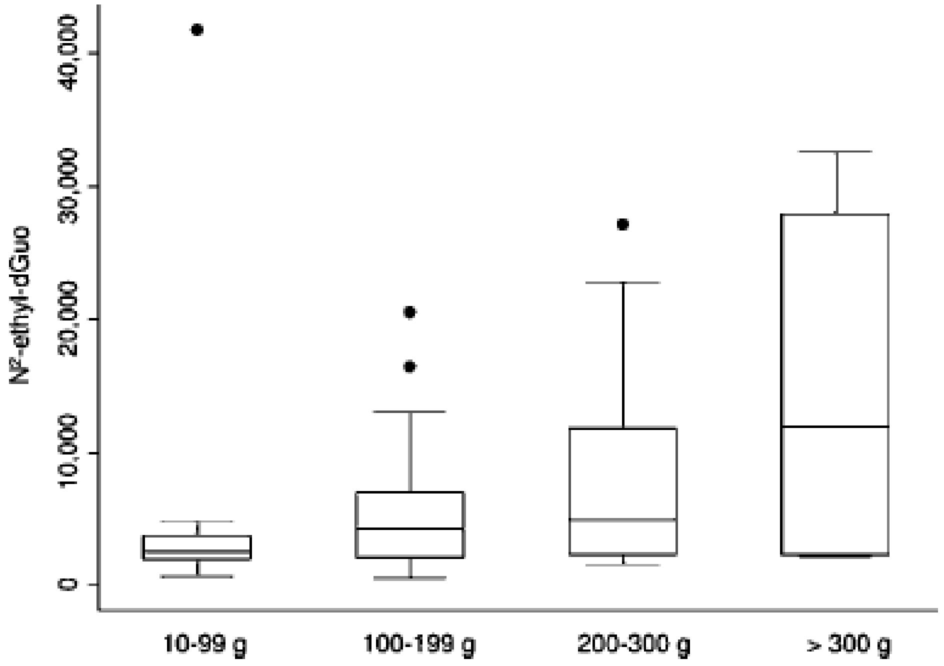




Association between acetaldehyde-derived DNA adducts and alcohol consumption

Levels of N^2-ethylidene-dG in DNA samples from two case-control studies conducted in Europe focusing on liver and head and neck cancers.

Peripheral blood samples from 50 non-drinkers and 50 drinkers consuming various amounts of alcohol (all smokers).



- High variability
- No information on the time between sample collection and consumption of the last drink

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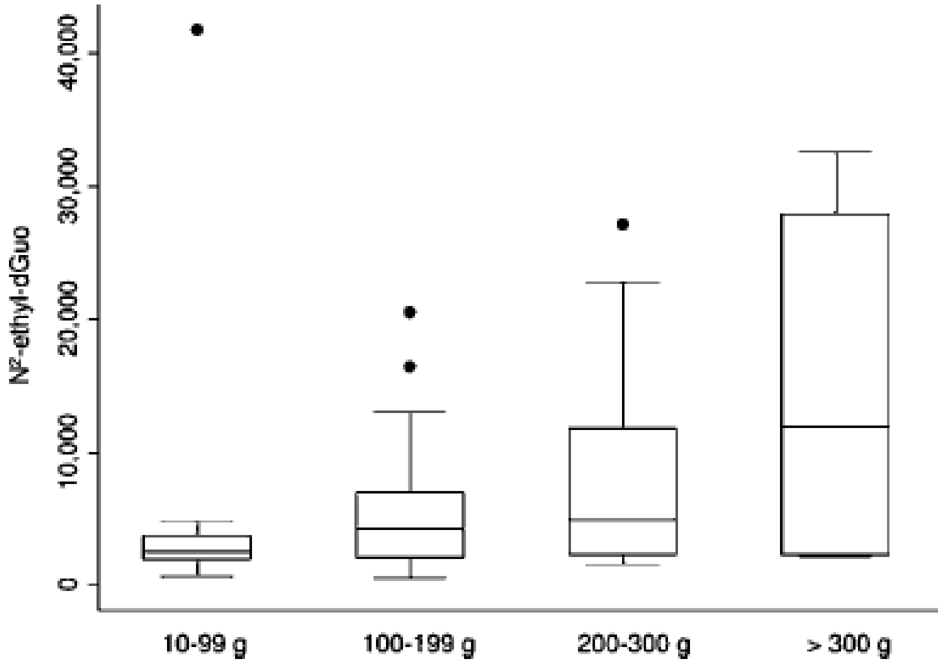

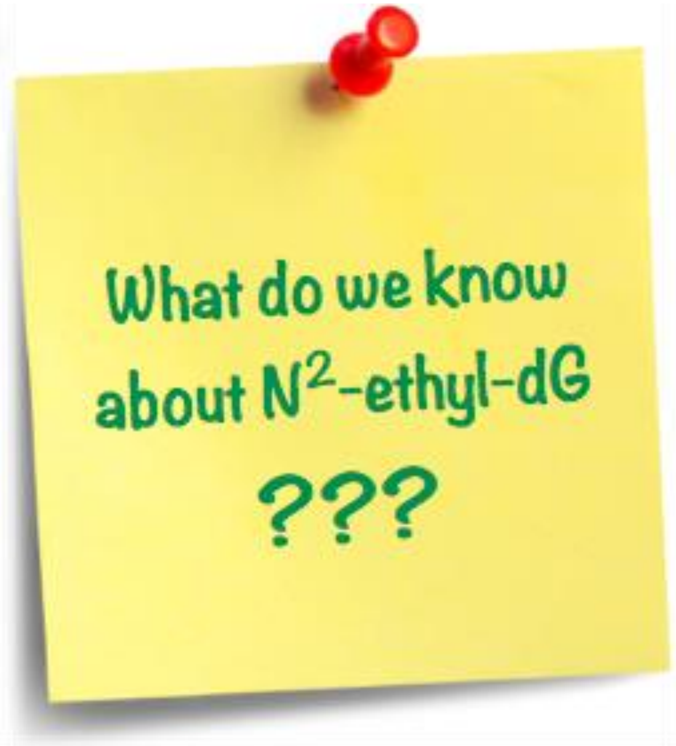

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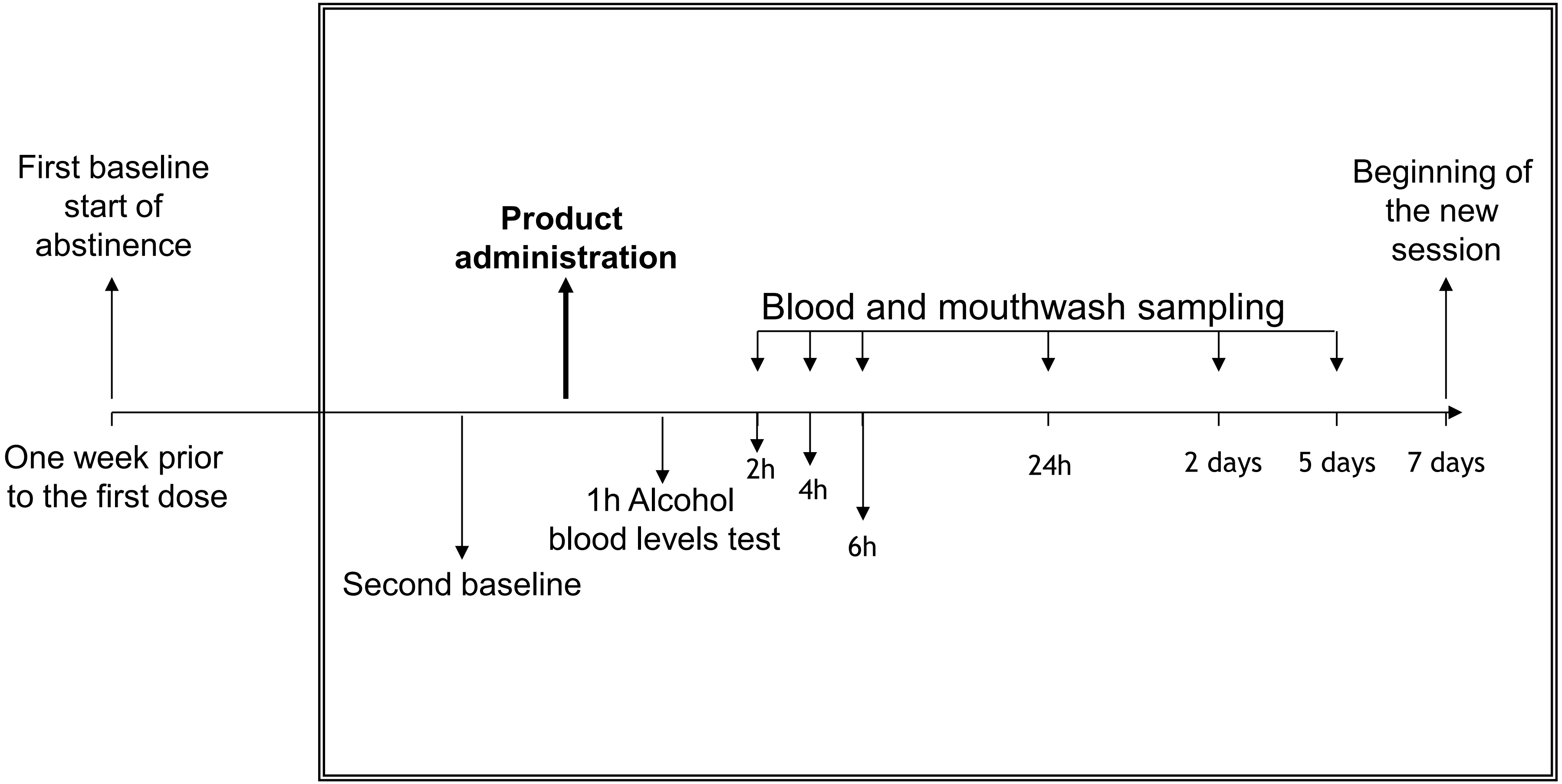
A study to Better Characterize the meaning of *N*^{2}-ethyl-dG



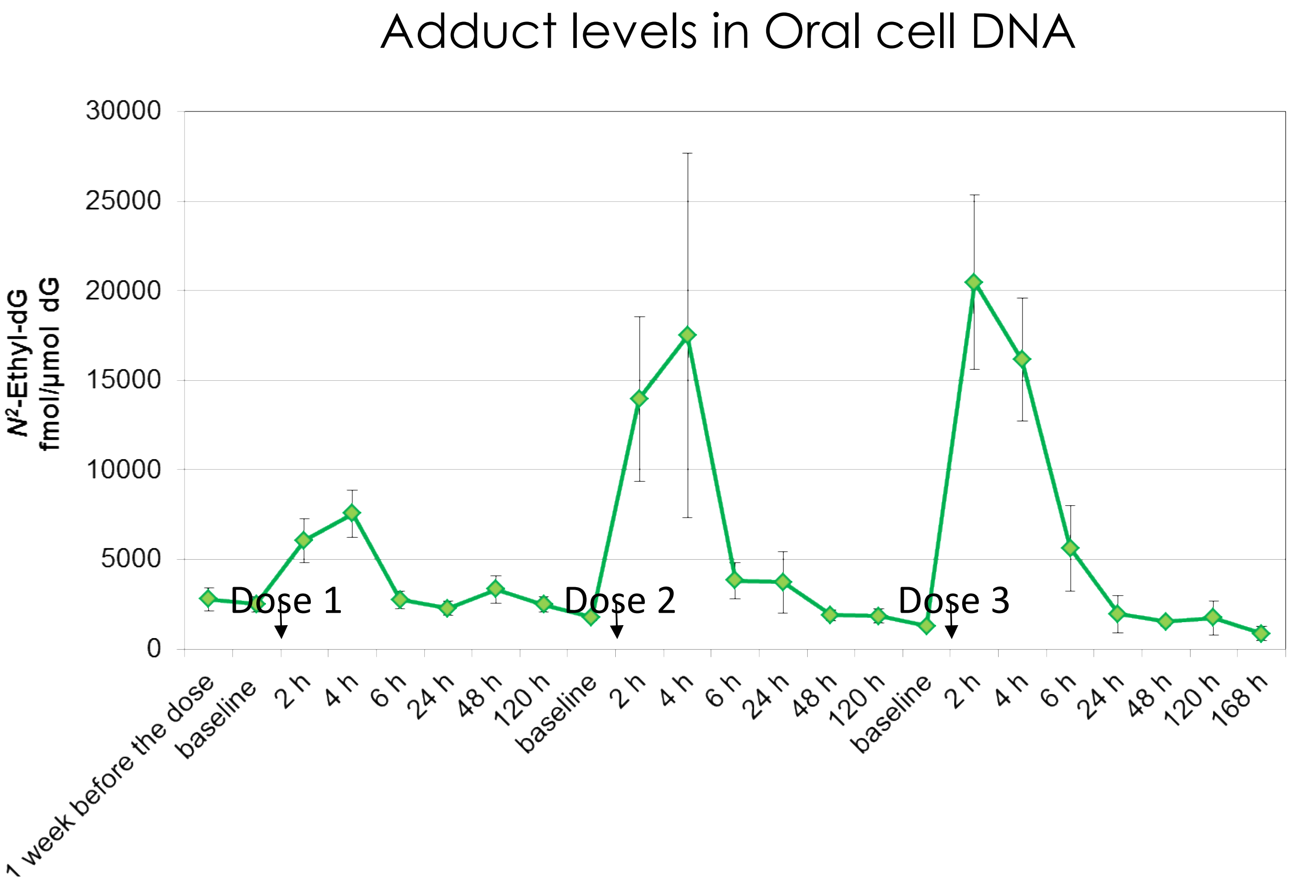
Repeated 3 times

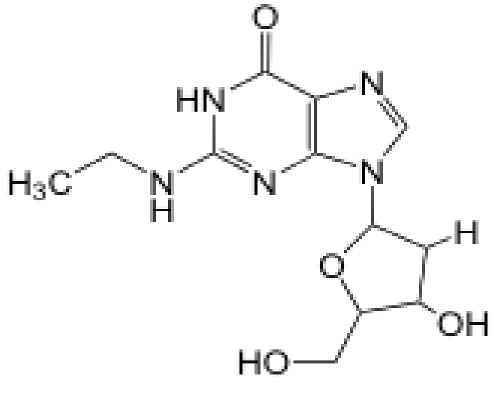
The alcohol dose

- Ethanol was administered using vodka 100 proof mixed with tonic water and with a little lime juice.
- The alcohol dose was calculated on the bases of gender and weight.
- The blood alcohol levels targeted were 0.03, 0.05 and 0.07%

	dose 1		dose 2		dose 3	
women	g of ethanol	BAL[a]	g of ethanol	BAL	g of ethanol	BAL
mean ± SD	19.4 ± 3.6	0.03 ± 0.02	28.0 ± 5.2	0.05 ± 0.01	36.6 ± 6.8	0.06 ± 0.01
men						
mean ± SD	27.2 ± 1.6	0.03 ± 0.01	39.3 ± 2.4	0.05 ± 0.01	51.4 ± 3.1	0.07 ± 0.01

Acetaldehyde reacts with DNA

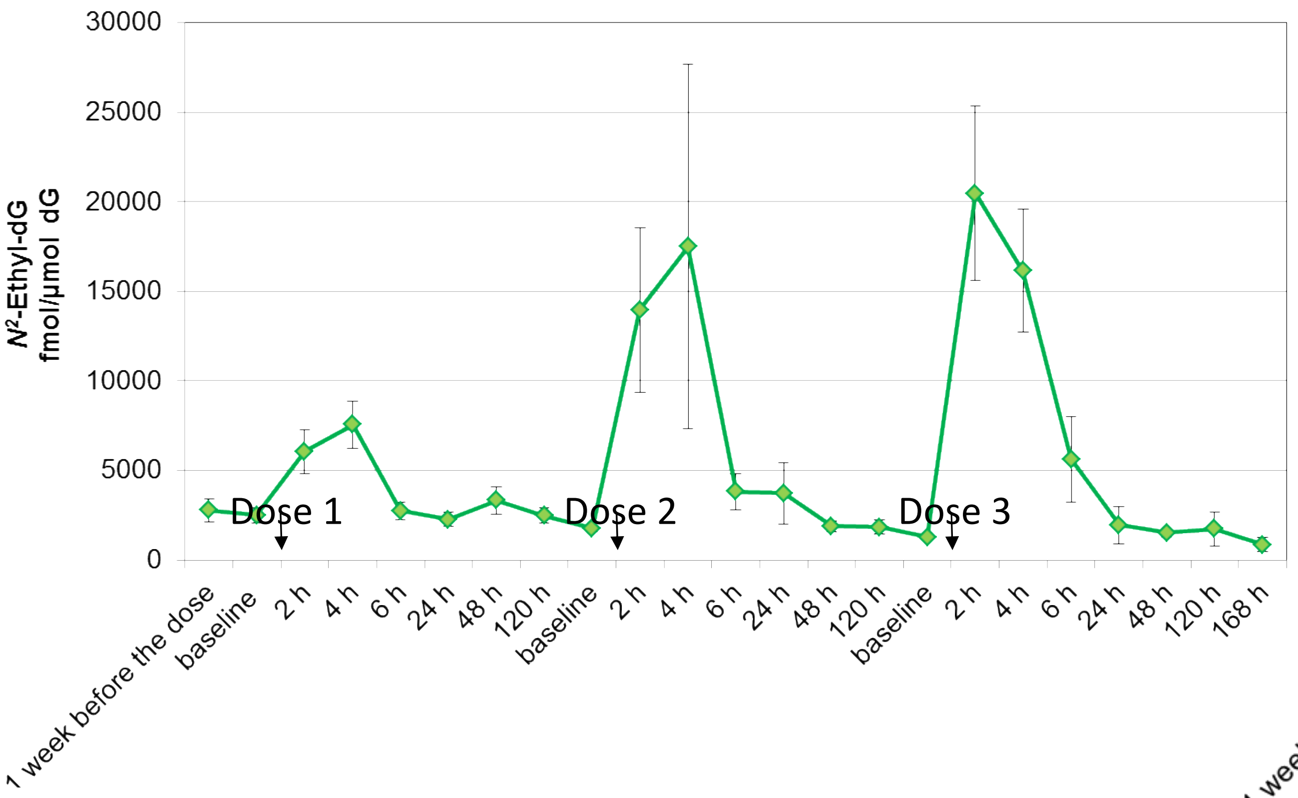




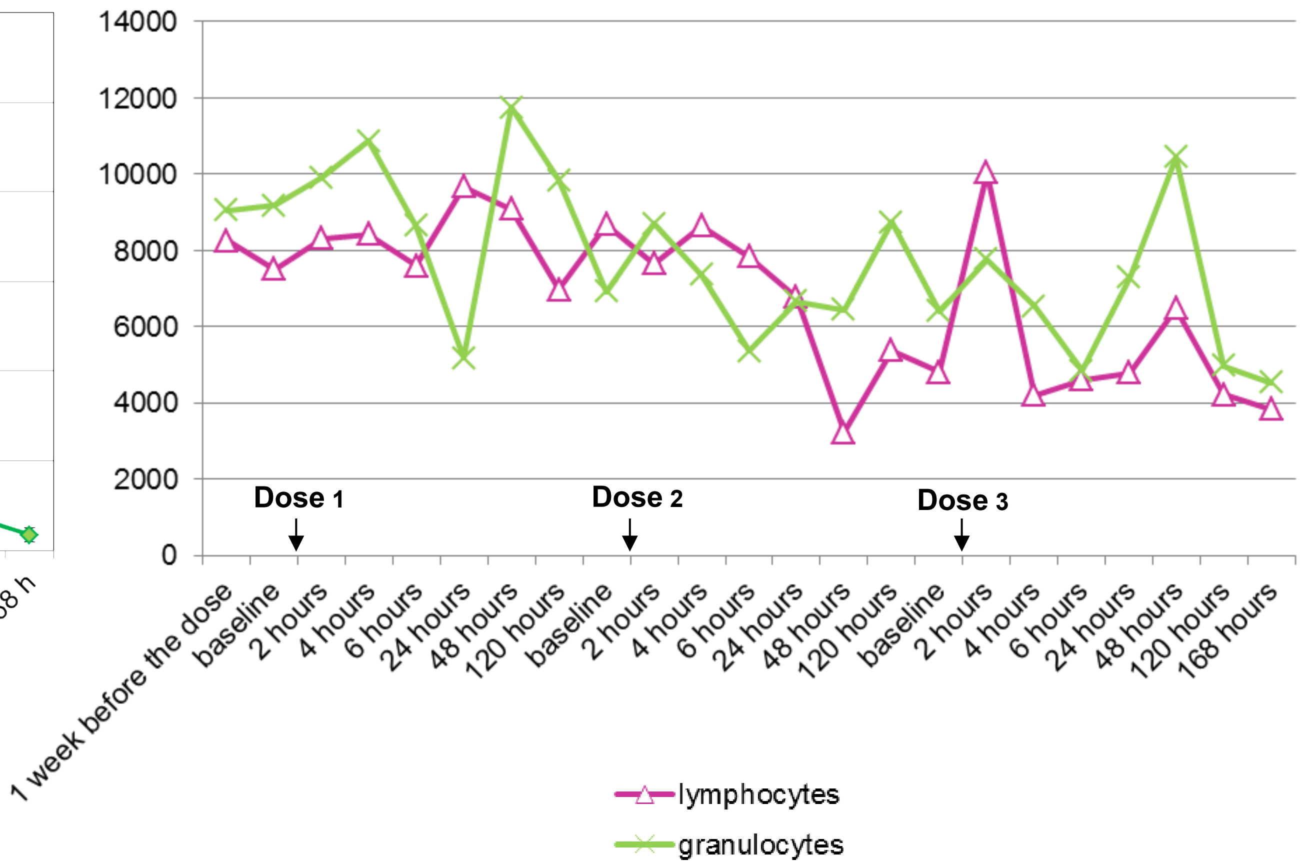
N^2-ethyl-dG

Acetaldehyde reacts with DNA

Adduct levels in Oral cell DNA

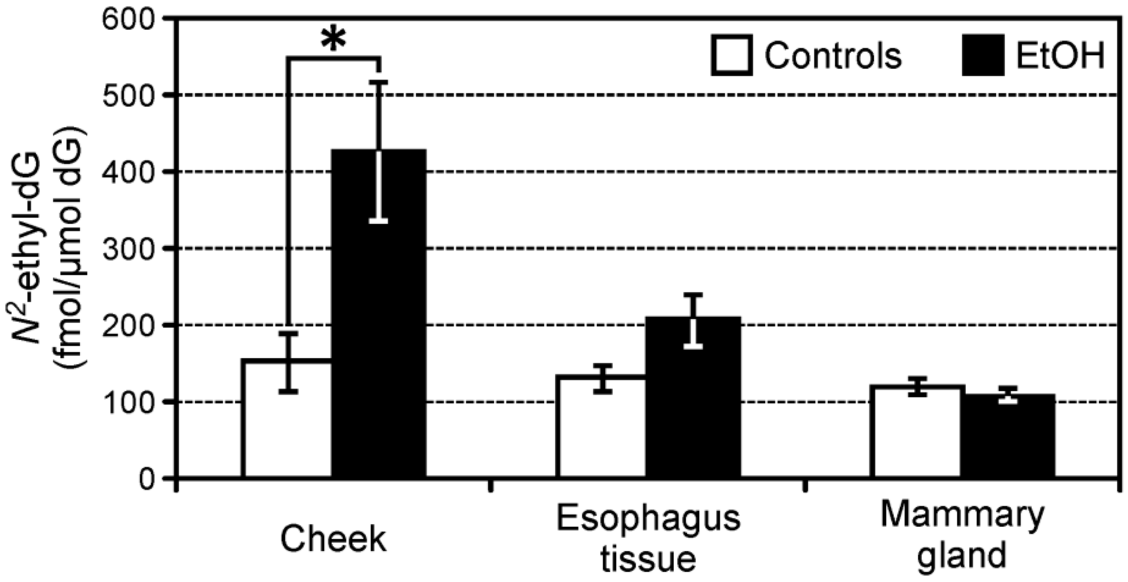


Adduct levels in Blood DNA



Acetaldehyde reacts with DNA

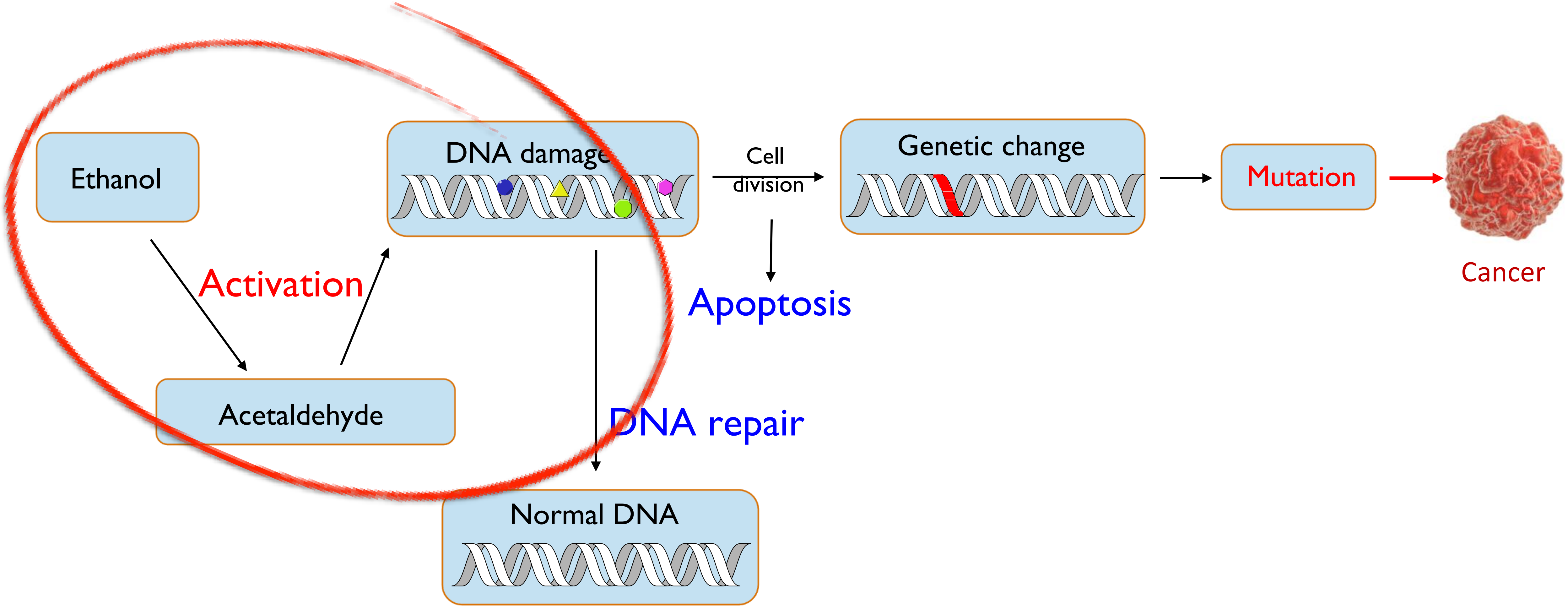
Adduct levels in tissues from Rhesus monkeys



Dr. Kathleen Grant
Oregon National Primate Research Center
Oregon Health & Science University

Acetaldehyde from alcohol consumption is a human carcinogen, Group 1

Acetaldehyde-derived DNA damage



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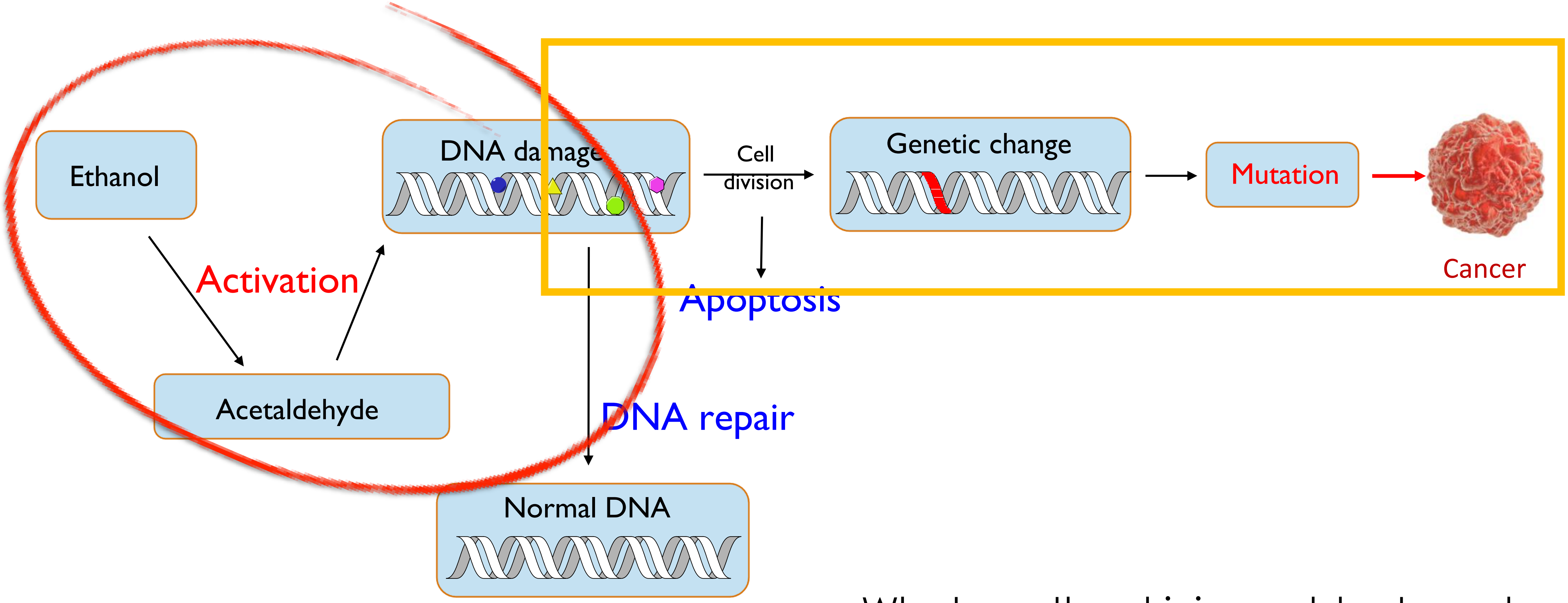


What are the driving adducts and what is their mutational impact?

SBS16 has been associated with alcohol consumption

nature genetics

ARTICLES

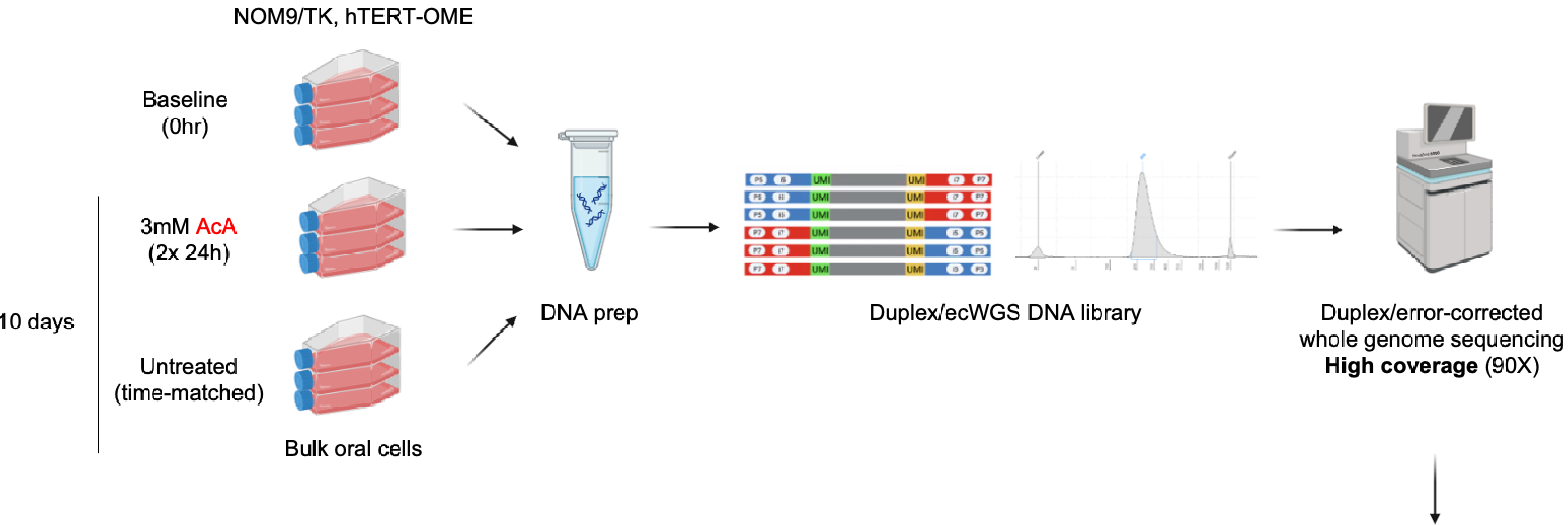
https://doi.org/10.1038/s41588-021-00928-6

Check for updates

Mutational signatures in esophageal squamous cell carcinoma from eight countries with varying incidence

Sarah Moody[1,25], Sergey Senkin[2,25], S. M. Ashiqul Islam[3,4,5], Jingwei Wang[1], Dariush Nasrollahzadeh[2,6], Ricardo Cortez Cardoso Penha[2], Stephen Fitzgerald[1], Erik N. Bergstrom[3,4,5], Joshua Atkins[2], Yudou He[3,4,5], Azhar Khandekar[3,4,5], Karl Smith-Byrne[2], Christine Carreira[7], Valerie Gaborieau[2], Calli Latimer[1], Emily Thomas[1], Irina Abnizova[1], Pauline E. Bucciarelli[1], David Jones[1], Jon W. Teague[1], Behnoush Abedi-Ardekani[2], Stefano Serra[8], Jean-Yves Scoazec[9], Hiva Saffar[10], Farid Azmoudeh-Ardalan[11], Masoud Sotoudeh[6], Arash Nikmanesh[6], Hossein Poustchi[6], Ahmadreza Niavarani[6], Samad Gharavi[6], Michael Eden[12], Paul Richman[13], Lia S. Campos[14], Rebecca C. Fitzgerald[15], Luis Felipe Ribeiro[16], Sheila Coelho Soares-Lima[16], Charles Dzamalala[17], Blandina Theophil Mmbaga[18], Tatsuhiro Shibata[19], Diana Menya[20], Alisa M. Goldstein[21], Nan Hu[21], Reza Malekzadeh[6], Abdolreza Fazel[22], Valerie McCormack[23], James McKay[2], Sandra Perdomo[2], Ghislaine Scelo[2,24], Estelle Chanudet[2], Laura Humphreys[1], Ludmil B. Alexandrov[3,4,5], Paul Brennan[2] and Michael R. Stratton[1] ✉

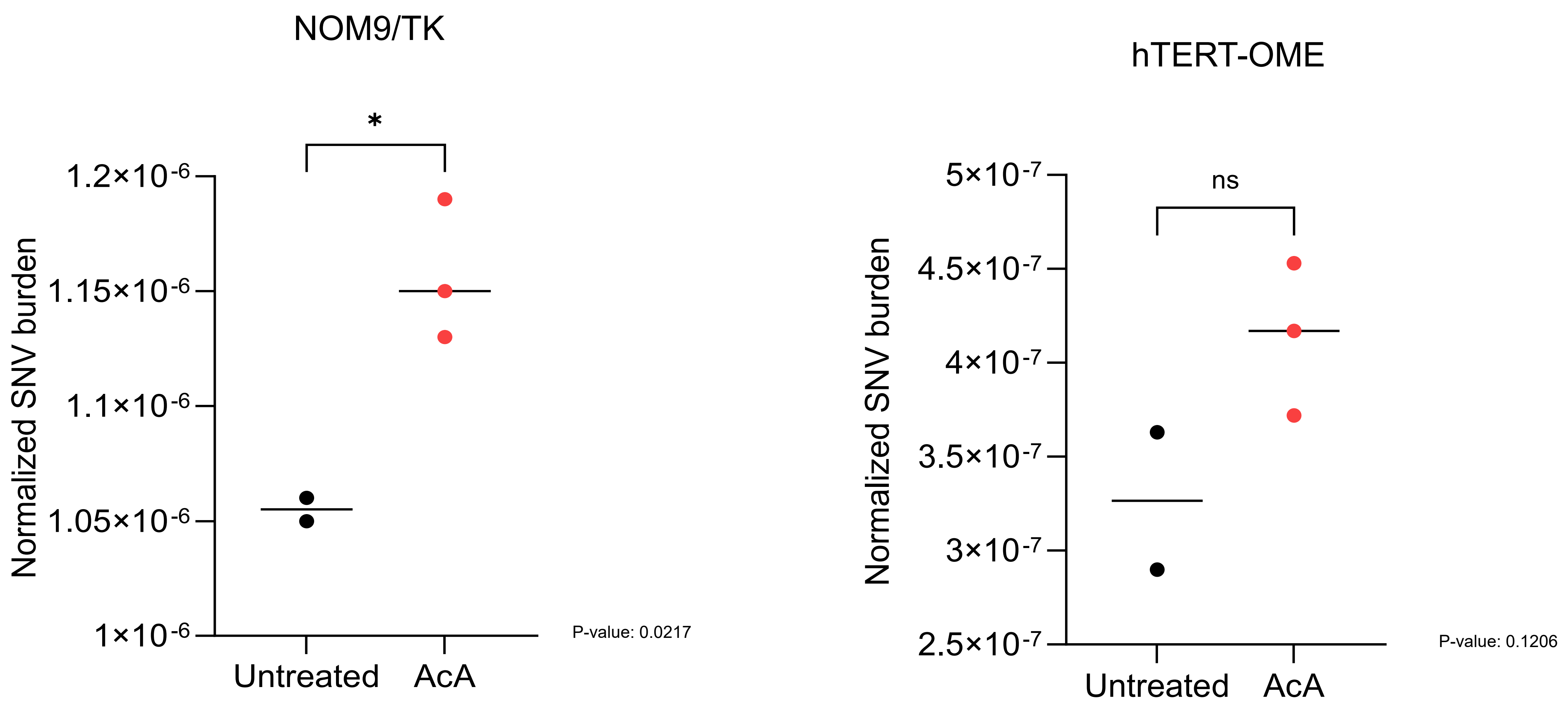
Integrating DNA adducts with mutations



Mutational signature analysis
SBS
DBS
Indels

In collaboration with Dr Jiri Zavadil IARC and Dr Ludmil Alexandrov UCSD

Mutation burden and Single Base Substitutions



Treatment-specific increase in overall mutation burden in NOM9/TK

Questions to focus on:

- monitor effects at low levels, and over time
- Identify driver DNA modifications
- Identify mechanisms following DNA damage
- using –omics approaches to monitor broadly and more granularly for changes induced by alcohol and dive deeper into the mechanisms of carcinogenesis
- investigate effects of exposure and cessation

Questions?

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Alcohol and Breast Cancer

- 7-10% increased risk for each drink/day in both pre- and post menopausal women

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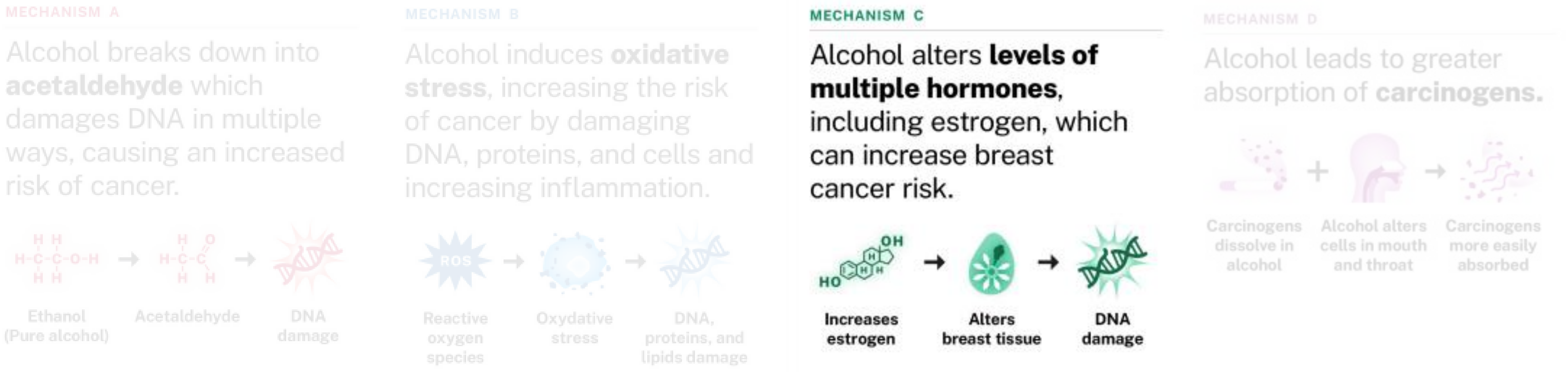
- 4-10% of breast cancers in USA are attributable to alcohol consumption, resulting in 9000-23000 new invasive breast cancer cases each year

Alcohol and Breast Cancer

- 7-10% increased risk for each drink/day in both pre- and post menopausal women
- 4-10% of breast cancers in USA are attributable to alcohol consumption, resulting in 9000-23000 new invasive breast cancer cases each year
- exposure during adolescence and early adulthood is more important than later in life, with alcohol consumption before first pregnancy being particularly important due to morphologic changes in the breast

Mechanisms of Cancer Formation



Alcohol effects on hormones

- Alcohol increases circulating levels of estrogen through reduced steroid degradation or increased aromatase activity

- To a lesser extent, Alcohol decreases progesterone

- Alcohol decreases testosterone in men and increases it in women

Hormonal imbalance mechanisms

Alcohol may influence breast cancer through ER-dependent pathways by increasing ER transcriptional activity, migration, and proliferation

In fact, alcohol is strongly associated with hormone receptor-positive breast cancers

Interestingly, sex and stress hormones influence alcohol consumption

Questions to focus on:

- monitor effects at low levels, and over time

- using –omics approaches to monitor broadly and more granularly for changes induced by alcohol and dive deeper into the mechanisms of carcinogenesis

- focus on sex differences and sex specific variables

- investigate effects of exposure and cessation

Acknowledgements

Kathleen Dubberley

Alcohol Epidemiologist Substance Misuse Prevention Section

Minnesota Department of Health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